

Off-Grid Automated Sustainable Irrigation Technical Design Report

1. Title and Basic Information

a. Project Title

- i. Off Grid Automated Sustainable Irrigation

b. Author Info

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c. Tagline

- i. An off-grid, water-cycled, automated irrigation system designed to confront the drought seasons in rural Illinois

2. Problem Statement and Engineering Objectives

a. Problem Statement

- i. The rainfall in Illinois is unevenly distributed throughout the year, even though agricultural production highly relies on natural precipitation. During peak summer months, rising temperatures drive crop water demand to its highest point, creating severe operational challenges during frequent drought periods.

b. Objectives

- i. Zero Grid Dependency: Operational 24/7 utilizing a self-sustaining solar/battery setup
- ii. Autonomous Irrigation: Continuous soil moisture evaluation without manual human intervention
- iii. Hydrological Efficiency: Recycling excess water back into the reservoir via underground collection pipe and natural stone filtration

- iv. Environmental Resilience: Physical protection rated against IP65 intrusion, wind lift, and freeze expansion damage

3. Hardware Bill of Materials (BOM)

Category	Component	Description	Quantity	Function and Engineering Value
Drainage & Irrigation	Blind Drain Pipe	Segtai Hdare 3.9" Permeable Corrugated Pipe	1	Buried underground to collect excess water for recycling. The synthetic polyester fiber jacket filter out dirt to prevent clogging
	Drip Line Kit	RAINPOINT 50ft Kit with adjustable Copper Nozzles	1	Delivers water directly to the plant
	HDPE Membrane	DGSL HDPE Pond Liner, 10*13ft Water Garden Liner	1	Prevent water loss into soil
	Water Reservoir	5-Gallon Heavy Duty Plastic Bucket	1	Acts as the central water reservoir to store collected water and house the submersible pump
Actuator and sensing	Water Pump	DIANN 12V Submersible Water Pump	1	Place inside the water well to pump water into the irrigation pipes
	Relay Module	AEDIKO 5V Relay Module	1	Acts as an automated

				switch: Arduino uses 5V signal to turn the 12V pump ON/OFF
	Soil Sensor	Songhe Capacitive Soil Moisture Sensor 3.3V to 5.5V	5	Reads soil moisture (capacitive design resist corrosion)
	Bread Board	ELEGOO 400 tie Points Breadboard	1	Provides a flexible platform to connect circuits and distribute 5V/GND power without soldering
Solar Power	Solar Panel	SOLPERK 20W 12V Solar Panel Kit	1	Collects sunlight to power the system and charge the battery
	Battery	ExpertPower 12V 7Ah battery SLA	1	Energy storage to ensure power during night and cloudy days
	Charge Controller	JNJstore 12V/24V 10A PWM Controller with LCD and USB	1	Protects the battery from overcharging and gives 5V power output to Arduino
	Extension Cable	Mygatti 14 Gauge Wire (2-Pack Red/Black)	1	Handle heavy DC power transmission between solar panel, controller, batter, pump, and relay

Protection	Enclosure	Raculety IP65 Waterproof Electronic Junction Box Power Enclosure (200*120*75mm)	1	Keeps the Arduino, relay, and wires safe from rain, snow and dust
		Foam Box	1	Provides thermal insulation for the battery and adds a secondary protective layer against freezing temperatures
	Safety Fuse	Anyongora 12V Inline Waterproof Fuse Holder with standard Fuses	4	Wired to battery positive terminal to prevent short-circuit fires
	Extension Cable	ENERJOUR 22AWG 3-Conductor Jacketed Cable	1	3-wire cable (5V/GND/Signal) prevents long-distance signal loss for connecting the sensor
	Connection Tubing	ALEKOR 3:1 Heat Shrink Tubing Kit	1	Internal glue melts upon heating to create an airtight, waterproof seal on wire joints
	Fixation/Tools	Electrical Tape, Heavy Duty Cable Ties, Tent Stakes, lighter, Wire Stripper	Multi	Used for wire routing, strain relief, waterproof sealing, and structural

				ground anchoring
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4. System Architecture and Wiring

The system consists of a 12V solar Power Bus and a 12V Logic Bus:

a. Power and Switch Loop (12V High Current Bus)

- i. Battery to Controller: Battery (+/-) — Charge Controller BAT (+/-) (Must be connected FIRST during assembly)
- ii. Solar panel to Controller: Solar Panel (+/-) — Charge Controller SOLAR (+/-)
- iii. Overcurrent Protection: Connect an Inline Fuse Holder (10A) in series on the red wire, immediately adjacent to the battery positive terminal
- iv. Pump Control Loop:
 - Charge Controller LOAD + — Relay COM (Common)
 - Relay NO (Normally Open) — Pump Positive Wire (Red)
 - Charge Controller LOAD - — Pump Negative Wire (Black, direct connection)

b. Sensing and Control Loop (5V Low-Voltage Logic)

- i. Arduino Power: Connect Arduino to the Charge Controller's 5V USB port via a standard USB cable
- ii. Relay to Arduino:
 - Arduino 5V — Relay VCC
 - Arduino GND — Relay GND
 - Arduino Pin 7 — Relay IN (Control Signal)
- iii. Soil Sensor to Arduino (via 22AWG 3C Extension Cable):
 - Arduino 5V — Sensor VCC (Red)

- Arduino GND — Sensor GND (Black)
- Arduino — Sensor AOUT (Yellow)

5. Power Logic and Solar Budget Calculation (pending benchtop validation)

6. Subsurface Drainage and Hydrological Recycling system (design confirmed / drafting)

7. Environmental and Disaster Defense

To ensure reliable, 24/7 autonomous operation under outdoor conditions in rural Illinois, the system incorporates physical defense mechanisms against high winds, heavy rain, and sub-zero temperatures

a. Wind and Storm Mitigation

- i. **Ballast Anchoring:** The main IP65 box and 12V battery are secured to a heavy concrete block using cable ties to prevent high winds from flipping the control unit
- ii. **Ground-Anchored Solar Panel:** The 20W solar panel is mounted at a low angle and pinned to the soil using metal tent stakes to reduce wind resistance

b. Rain and Moisture Defense

- i. **Sealed Junction Box:** The IP65 enclosure is placed on a stable surface, with all wire entries sealed using hot melt glue to block rainwater and moisture
- ii. **Sensor Protection:** Sensor joints are sealed using 3:1 adhesive-lined heat shrink tubing to prevent short circuits when the soil is saturated

c. Sub-Zero Winterization Protocol

- i. **Battery Thermal Insulation:** The 12V battery is housed inside a foam box to prevent power loss in freezing weather

- ii. Line Drainage: Before winter freeze, the pump is run briefly outside the reservoir to clear residual water from the drip lines and prevent tube bursting

8. Software Logic and Algorithm

The system runs on an Arduino UNO to control the 12V water pump based on moisture. The software is split into two parts: a quick calibration script to find baseline numbers, and a main program that runs the automatic watering loop. The complete code files can be found on my GitHub repository.

a. Sensor Calibration Protocol and Threshold Derivation

- i. Before running the main automatic control program, a separate calibration script is uploaded to the Arduino UNO to test the capacitive soil moisture sensor. This process establishes two physical boundary baselines from the sensor's raw analog output:
 - 1. Air Value: The reading when the sensor is completely dry in the air (representing 0% soil moisture)
 - 2. Water Value: The reading when the sensor is placed in water (representing 100% soil saturation)
- ii. Because real-world soil does not operate at these absolute physical extremes, a ~30% safety margin is applied to these threshold limits to calculate practical operational setpoints:
 - 1. Dry Threshold: Calculated at a ~30% offset from the dry value to trigger the pump when soil is dehydrated
 - 2. Wet Threshold: Calculated at a ~30% offset from the wet value to turn off the pump and prevent over-watering or root rot

b. Main Control Logic and Implementation

To control the irrigation system safely and efficiently, the main Arduino program combines three key logical features into a single non-blocking execution loop:

- i. Dual-Threshold Hysteresis Logic: Setting only a single moisture threshold causes the relay to toggle ON and OFF extremely fast when readings sit right on the boundary (“relay chattering”). This causes severe wear on the relay and can easily burn out the pump motor. By using two separate thresholds with a buffer zone in between, the system avoids unstable switching:
 1. Dry Threshold (*dry threshold* = 700): If the reading exceeds 700 (soil is dry), Pin 7 outputs LOW to turn ON the pump
 2. Wet Threshold (*wet threshold* = 500): The pump stays running until the moisture reading drops below 500 (soil is wet), the Pin 7 outputs HIGH to turn OFF the pump
 3. Buffer Zone: When reading stay between 500 and 700, the system holds its previous state
- ii. Water Soil Absorption Interval: The reading interval (*loopDelay* = 60000) gives pumped water enough time to seep into the soil and reach the sensor, prevent over-watering caused by reading too frequently before moisture disperses
- iii. *millis()* Non-Blocking Timing and Overflow Protection: Standard *delay()* is avoided because it completely freezes the processor. Using *millis()* keeps the board active. Furthermore, calculating time with subtraction (*currentMillis* - *previousMillis* >= *interval*) prevents system crashes when Arduino's 32-bit millisecond timer overflows and resets to zero after ~49.7 days of continuous operation

Note: The threshold values (dry threshold = 700 & wet threshold = 500) serve as initial system approximation, these will be updated based on the result of calibration

9. Testing and Results (pending benchtop calibration & field data verification)

10. Deployment Guide and Future Work (In progress / drafting)

11. AI Assistance Declaration

Large Language Models (ChatGPT / Google Gemini) were utilized solely for grammar correction, language polishing, and document structure formatting during the drafting of this report. All underlying hardware designs, experimental testing, data collection, and software logic were independently developed and validated by the author